

ECON2050 Tutorial 7

Convexity

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Today's Plan

- ▶ Announcement: Assignment 2 has been released.
 - ▶ Due date: 1st October 16:00pm.
- ▶ A1 is still being marked. Hopefully, we can finish before A2 is due.
- ▶ Tutorial questions: Q1–Q5

Tutorial Question 5

Problem 5

Prove from the definition of a strictly convex function, and without using any theorem about convex functions, that

$$f(x) = x^2$$

is strictly convex.

Recap: Convex functions

Let $S \subseteq \mathbb{R}^n$ be convex and $f: S \rightarrow \mathbb{R}$.

Convexity

f is **convex** if, for all $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$,

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

Strict convexity

f is **strictly convex** if the inequality is strict whenever $\mathbf{x} \neq \mathbf{y}$ and $\lambda \in (0, 1)$.

Problem 5: Knowledge used

Knowledge used

A function f is strictly convex if, for all $x', x'' \in \mathbb{R}$ with $x' \neq x''$ and all $\lambda \in (0, 1)$,

$$f(\lambda x' + (1 - \lambda)x'') < \lambda f(x') + (1 - \lambda)f(x'').$$

The proof must establish this inequality directly from the definition.

Problem 5: Expand both sides

Take $x', x'' \in \mathbb{R}$ with $x' \neq x''$ and $\lambda \in (0, 1)$. For $f(x) = x^2$,

$$\begin{aligned} f(\lambda x' + (1 - \lambda)x'') &= (\lambda x' + (1 - \lambda)x'')^2 \\ &= \lambda^2(x')^2 + 2\lambda(1 - \lambda)x'x'' + (1 - \lambda)^2(x'')^2, \end{aligned}$$

while

$$\lambda f(x') + (1 - \lambda)f(x'') = \lambda(x')^2 + (1 - \lambda)(x'')^2.$$

Problem 5: Compare the two expressions

Consider the difference:

$$\begin{aligned} & \lambda f(x') + (1 - \lambda)f(x'') - f(\lambda x' + (1 - \lambda)x'') \\ &= \lambda(1 - \lambda)(x')^2 - 2\lambda(1 - \lambda)x'x'' + \lambda(1 - \lambda)(x'')^2 \\ &= \lambda(1 - \lambda)[(x')^2 - 2x'x'' + (x'')^2] \\ &= \lambda(1 - \lambda)(x' - x'')^2. \end{aligned}$$

Because

$$0 < \lambda < 1 \implies \lambda(1 - \lambda) > 0,$$

and because $x' \neq x''$,

$$(x' - x'')^2 > 0.$$

Hence

$$\lambda(1 - \lambda)(x' - x'')^2 > 0.$$

Therefore,

$$f(\lambda x' + (1 - \lambda)x'') < \lambda f(x') + (1 - \lambda)f(x''),$$

so $f(x) = x^2$ is **strictly convex**.

Tutorial Question 1

Problem 1

Show that if $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ is a convex function and $b \in \mathbb{R}$, then

$$S = \{(x, y) \in \mathbb{R}^2 : g(x, y) \leq b\}$$

is a convex set.

Recap: Convex objects

Convex sets

A set $S \subseteq \mathbb{R}^n$ is **convex** if for every two **vectors** $\mathbf{x}, \mathbf{y} \in S$, the entire line segment between them is contained in S :

$$[\mathbf{x}, \mathbf{y}] \subseteq S.$$

Equivalently,

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in S \quad \text{for all } \mathbf{x}, \mathbf{y} \in S, \lambda \in [0, 1].$$

Let $S \subseteq \mathbb{R}^n$ be convex and $f: S \rightarrow \mathbb{R}$.

Convexity function

f is **convex** if, for all $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$,

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

f is **strictly convex** if the inequality is strict whenever $\mathbf{x} \neq \mathbf{y}$ and $\lambda \in (0, 1)$.

Problem 1: Knowledge used

Knowledge used

- ▶ Definition of a convex set: every convex combination of two points in the set must remain in the set.
- ▶ Definition of a convex function:

$$g(\lambda \mathbf{u} + (1 - \lambda) \mathbf{v}) \leq \lambda g(\mathbf{u}) + (1 - \lambda) g(\mathbf{v}).$$

Problem 1: Start from two points in S

Take $(x, y), (x', y') \in S$ and $\lambda \in [0, 1]$.

Because the two points belong to S ,

$$g(x, y) \leq b, \quad g(x', y') \leq b.$$

Multiplying by the non-negative weights gives

$$\lambda g(x, y) \leq \lambda b, \quad (1 - \lambda)g(x', y') \leq (1 - \lambda)b.$$

By convexity of g ,

$$\begin{aligned} g(\lambda(x, y) + (1 - \lambda)(x', y')) &\leq \lambda g(x, y) + (1 - \lambda)g(x', y') \\ &\leq \lambda b + (1 - \lambda)b \\ &= b. \end{aligned}$$

Therefore,

$$\lambda(x, y) + (1 - \lambda)(x', y') \in S.$$

Hence S is **convex**.

Tutorial Question 2

Problem 2

Suppose $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ can be written as

$$f(x, y) = g(x) + h(y),$$

where both g and h are convex functions of one variable. Show that f is also convex.

Recap: Convex functions

Let $S \subseteq \mathbb{R}^n$ be convex and $f: S \rightarrow \mathbb{R}$.

Convexity

f is **convex** if, for all $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$,

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

Strict convexity

f is **strictly convex** if the inequality is strict whenever $\mathbf{x} \neq \mathbf{y}$ and $\lambda \in (0, 1)$.

Problem 2: Knowledge used

Knowledge used

- ▶ Definition of convexity for a function on $\mathbb{R}^2$.
- ▶ Convexity inequalities for the one-variable functions g and h .
- ▶ Add the two convexity inequalities and regroup the terms.

Problem 2: Evaluate f at a convex combination

Take $(x', y'), (x'', y'') \in \mathbb{R}^2$ and $\lambda \in [0, 1]$. Then

$$\begin{aligned} & f(\lambda(x', y') + (1 - \lambda)(x'', y'')) \\ &= f(\lambda x' + (1 - \lambda)x'', \lambda y' + (1 - \lambda)y'') \\ &= g(\lambda x' + (1 - \lambda)x'') + h(\lambda y' + (1 - \lambda)y''). \end{aligned}$$

Because g and h are convex,

$$\begin{aligned} g(\lambda x' + (1 - \lambda)x'') &\leq \lambda g(x') + (1 - \lambda)g(x''), \\ h(\lambda y' + (1 - \lambda)y'') &\leq \lambda h(y') + (1 - \lambda)h(y''). \end{aligned}$$

Adding,

$$\begin{aligned} f(\lambda(x', y') + (1 - \lambda)(x'', y'')) &\leq \lambda[g(x') + h(y')] \\ &\quad + (1 - \lambda)[g(x'') + h(y'')] \\ &= \lambda f(x', y') + (1 - \lambda)f(x'', y''). \end{aligned}$$

Thus, f is **convex**.

Tutorial Question 3

Problem 3

Show that the following sets are convex.

1.

$$S = \{(x, y) \in \mathbb{R}^2 : y \geq x, x^2 + y^2 \leq 4\}.$$

2.

$$T = \{(x, y) \in \mathbb{R}^2 : y \geq x^2, y \leq 4\}.$$

Recap: Intersections and unions

Intersection property

Every intersection of convex sets is convex.

If each S_i is convex, then $\bigcap_{i \in I} S_i$ is convex.

Warning

In general, a union of convex sets need not be convex.

Recap: Operations preserving convexity and concavity

Non-negative linear combinations

If $f_1, \dots, f_k$ are convex [concave] on a convex set S , and $\alpha_i \geq 0$, then

$$\alpha_1 f_1 + \dots + \alpha_k f_k$$

is also convex [concave].

Negation and affine functions

If f is convex [concave], then $-f$ is concave [convex]. Every affine function is both convex and concave.

Recap: Epigraph and hypograph

Let $S \subseteq \mathbb{R}^n$ be convex and $f: S \rightarrow \mathbb{R}$.

Epigraph

$$\text{epi}(f) = \{(\mathbf{x}, y) \in S \times \mathbb{R} : y \geq f(\mathbf{x})\}.$$

Hypograph

$$\text{hyp}(f) = \{(\mathbf{x}, y) \in S \times \mathbb{R} : y \leq f(\mathbf{x})\}.$$

Epigraph/hypograph characterization

Let $S \subseteq \mathbb{R}^n$ be convex and $f: S \rightarrow \mathbb{R}$.

- ▶ f is convex iff $\text{epi}(f)$ is a convex set.
- ▶ f is concave iff $\text{hyp}(f)$ is a convex set.

Problem 3: Knowledge used

Knowledge used

- ▶ The epigraph of a convex function is a convex set.
- ▶ From Problem 1: a lower contour set $\{\mathbf{x} : g(\mathbf{x}) \leq b\}$ of a convex function is convex.
- ▶ Every affine function is convex.
- ▶ x^2 and $x^2 + y^2$ are convex.
- ▶ The intersection of convex sets is convex.

Problem 3(1): Decompose S

Write

$$S = S_1 \cap S_2,$$

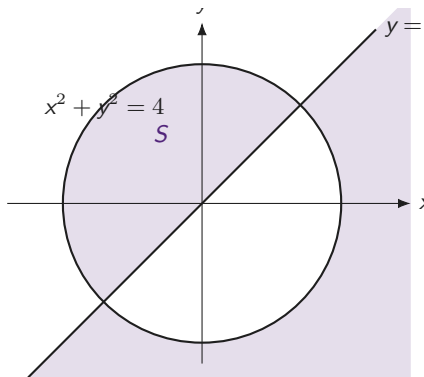
where

$$S_1 = \{(x, y) : y \geq x\}, \quad S_2 = \{(x, y) : x^2 + y^2 \leq 4\}.$$

- ▶ $S_1 = \text{epi}(f_1)$ with $f_1(x) = x$. Since f_1 is affine, it is convex; hence S_1 is convex.
- ▶ S_2 is a lower contour set of $f_2(x, y) = x^2 + y^2$, which is convex. Hence, by Problem 1, S_2 is convex.

Therefore, since $S = S_1 \cap S_2$, **S is convex.**

Problem 3(1): Graphical view



The feasible region is the part of the disk $x^2 + y^2 \leq 4$ lying above the line $y = x$.

Problem 3(2): Decompose T

Write

$$T = T_1 \cap T_2,$$

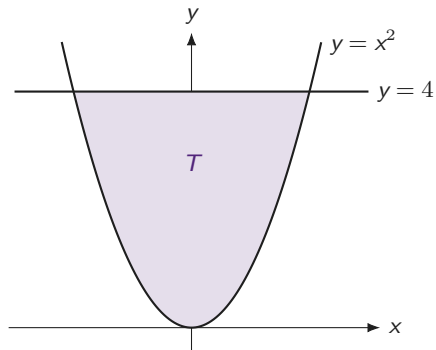
where

$$T_1 = \{(x, y) : y \geq x^2\}, \quad T_2 = \{(x, y) : y \leq 4\}.$$

- ▶ $T_1 = \text{epi}(g_1)$ with $g_1(x) = x^2$. Since g_1 is convex, T_1 is convex.
- ▶ $T_2 = \{(x, y) : g_2(x, y) \leq 4\}$ with $g_2(x, y) = y$. Since g_2 is affine and hence convex, Problem 1 implies that T_2 is convex.

Therefore, since $T = T_1 \cap T_2$, **T is convex.**

Problem 3(2): Graphical view



The feasible region lies above the parabola $y = x^2$ and below the line $y = 4$.

Tutorial Question 4

Problem 4

Analyze whether the following functions are concave or convex.

1. $f_1(x, y) = 4x^2 + e^{2y^2}$.
2. $f_2(x, y) = \ln(3x + 2y)$.

Recap: Monotonicity and derivatives

Monotonicity

For a differentiable $f: I \rightarrow \mathbb{R}$ on an open interval I :

- ▶ f is increasing iff $f' \geq 0$ on I ;
- ▶ $f' > 0$ on I implies f is strictly increasing;
- ▶ f is decreasing iff $f' \leq 0$ on I ;
- ▶ $f' < 0$ on I implies f is strictly decreasing.

Recap: One-variable convexity tests

First derivative

For differentiable $f: I \rightarrow \mathbb{R}$ on an open interval I :

- ▶ f is (strictly) convex iff f' is (strictly) increasing;
- ▶ f is (strictly) concave iff f' is (strictly) decreasing.

Second derivative

If f is twice differentiable,

$$f'' \geq 0 \iff f \text{ is convex}, \quad f'' > 0 \implies f \text{ is strictly convex},$$

$$f'' \leq 0 \iff f \text{ is concave}, \quad f'' < 0 \implies f \text{ is strictly concave}.$$

Recap: Operations preserving convexity and concavity

Non-negative linear combinations

If $f_1, \dots, f_k$ are convex [concave] on a convex set S , and $\alpha_i \geq 0$, then

$$\alpha_1 f_1 + \dots + \alpha_k f_k$$

is also convex [concave].

Negation and affine functions

If f is convex [concave], then $-f$ is concave [convex]. Every affine function is both convex and concave.

Composition rule

Let $S \subseteq \mathbb{R}^n$ be convex.

- ▶ If $f: S \rightarrow \mathbb{R}$ is convex and $g: \mathbb{R} \rightarrow \mathbb{R}$ is convex and increasing, then $g \circ f$ is convex.
- ▶ If $f: S \rightarrow \mathbb{R}$ is concave and $g: \mathbb{R} \rightarrow \mathbb{R}$ is concave and increasing, then $g \circ f$ is concave.

Problem 4: Knowledge used

Knowledge used

- ▶ x^2 is convex.
- ▶ e^u is increasing and convex.
- ▶ $\ln u$ is increasing and concave on its domain.
- ▶ Non-negative linear combinations of convex functions are convex.
- ▶ If the outer function is increasing, the relevant composition rule preserves convexity or concavity.
- ▶ The sum result proved in Problem 2.

Problem 4(1): $f_1(x, y) = 4x^2 + e^{2y^2}$

Write

$$f_1(x, y) = g_1(x) + h_1(y), \quad g_1(x) = 4x^2, \quad h_1(y) = e^{2y^2}.$$

- ▶ g_1 is convex because x^2 is convex and multiplication by $4 > 0$ preserves convexity.
- ▶ Write $h_1 = m \circ n$, where

$$m(u) = e^u, \quad n(y) = 2y^2.$$

Here m is increasing and convex, while n is convex. Hence h_1 is convex.

By Problem 2, f_1 is convex.

Problem 4(2): $f_2(x, y) = \ln(3x + 2y)$

Write

$$f_2 = h_2 \circ g_2, \quad h_2(u) = \ln u, \quad g_2(x, y) = 3x + 2y.$$

- ▶ $h_2(u) = \ln u$ is increasing and concave on its domain.
- ▶ $g_2(x, y) = 3x + 2y$ is linear, hence affine and therefore concave.

By the composition rule for an increasing concave outer function, f_2 is concave.